

Single studies using the CohortMethod package

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1 Introduction

This vignette describes how you can use the `CohortMethod` package to perform a single new-user cohort study. We will walk through all the steps needed to perform an exemplar study, and we have selected the well-studied topic of the effect of coxibs versus non-selective non-steroidal anti-inflammatory drugs (NSAIDs) on gastrointestinal (GI) bleeding-related hospitalization. For simplicity, we focus on one coxib – celecoxib – and one non-selective NSAID – diclofenac.

2 Data extraction

The first step in running the `CohortMethod` is extracting all necessary data from the database server holding the data in the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM) format.

2.1 Configuring the connection to the server

We need to tell R how to connect to the server where the data are. `CohortMethod` uses the `DatabaseConnector` package, which provides the `createConnectionDetails` function. Type `?createConnectionDetails` for the specific settings required for the various database management systems (DBMS). For example, one might connect to a PostgreSQL database using this code:

```
library(CohortMethod)
connectionDetails <- createConnectionDetails(dbms = "postgresql",
                                             server = "localhost/ohdsi",
                                             user = "joe",
                                             password = "supersecret")

cdmDatabaseSchema <- "my_cdm_data"
cohortDatabaseSchema <- "my_results"
cohortTable <- "my_cohorts"
options(sqlRenderTempEmulationSchema = NULL)
```

The last few lines define the `cdmDatabaseSchema`, `cohortDatabaseSchema`, and `cohortTable` variables. We'll use these later to tell R where the data in CDM format live, and where we want to write intermediate tables. Note that for Microsoft SQL Server, databaseschemas need to specify both the database and the schema, so for example `cdmDatabaseSchema <- "my_cdm_data.dbo"`. For database platforms that do not support temp tables, such as Oracle, it is also necessary to provide a schema where the user has write access that can be used to emulate temp tables. PostgreSQL supports temp tables, so we can set `options(sqlRenderTempEmulationSchema = NULL)` (or not set the `sqlRenderTempEmulationSchema` at all.)

2.2 Preparing the exposures and outcome(s)

We need to define the exposures and outcomes for our study. Here, we will define our exposures using the OHDSI `Capr` package. We define two exposure cohorts, one for celecoxib and one for diclofenac. It is often a good idea to restrict your analysis to a specific indication, to maximize the comparability of the two cohorts. In this case, we will restrict to osteoarthritis of the knee. We will create a cohort for this indication, starting at the first ever diagnosis, and ending at observation period end.

```
library(Capr)

celecoxibConceptId <- 1118084
diclofenacConceptId <- 1124300
osteoArthritisOfKneeConceptId <- 4079750

celecoxib <- cs(
  descendants(celecoxibConceptId),
  name = "Celecoxib"
)

celecoxibCohort <- cohort(
  entry = entry(
    drugExposure(celecoxib)
  ),
  exit = exit(endStrategy = drugExit(celecoxib,
```

```

                                persistenceWindow = 30,
                                surveillanceWindow = 0))
)

diclofenac <- cs(
  descendants(diclofenacConceptId),
  name = "Diclofenac"
)

diclofenacCohort <- cohort(
  entry = entry(
    drugExposure(diclofenac)
  ),
  exit = exit(endStrategy = drugExit(diclofenac,
                                    persistenceWindow = 30,
                                    surveillanceWindow = 0))
)

osteoArthritisOfKnee <- cs(
  descendants(osteoArthritisOfKneeConceptId),
  name = "Osteoarthritis of knee"
)

osteoArthritisOfKneeCohort <- cohort(
  entry = entry(
    conditionOccurrence(osteoArthritisOfKnee, firstOccurrence())
  ),
  exit = exit(
    endStrategy = observationExit()
  )
)
# Note: this will automatically assign cohort IDs 1,2, and 3, respectively:
exposuresAndIndicationCohorts <- makeCohortSet(celecoxibCohort,
                                              diclofenacCohort,
                                              osteoArthritisOfKneeCohort)

```

We'll pull the outcome definition from the OHDSI PhenotypeLibrary:

```

library(PhenotypeLibrary)
outcomeCohorts <- getPlCohortDefinitionSet(77) # GI bleed

```

We combine the exposure and outcome cohort definitions, and use CohortGenerator to generate the cohorts:

```

allCohorts <- bind_rows(outcomeCohorts,
                        exposuresAndIndicationCohorts)

library(CohortGenerator)
cohortTableNames <- getCohortTableNames(cohortTable = cohortTable)
createCohortTables(connectionDetails = connectionDetails,
                   cohortDatabaseSchema = cohortDatabaseSchema,
                   cohortTableNames = cohortTableNames)
generateCohortSet(connectionDetails = connectionDetails,
                  cdmDatabaseSchema = cdmDatabaseSchema,
                  cohortDatabaseSchema = cohortDatabaseSchema,
                  cohortTableNames = cohortTableNames,

```

```
cohortDefinitionSet = allCohorts)
```

If all went well, we now have a table with the cohorts of interest. We can see how many entries per cohort:

```
connection <- DatabaseConnector::connect(connectionDetails)
sql <- "SELECT cohort_definition_id, COUNT(*) AS count
FROM @cohortDatabaseSchema.@cohortTable
GROUP BY cohort_definition_id"
cohortCounts <- DatabaseConnector::renderTranslateQuerySql(
  connection = connection,
  sql = sql,
  cohortDatabaseSchema = cohortDatabaseSchema,
  cohortTable = cohortTable
)
DatabaseConnector::disconnect(connection)
```

```
## cohort_concept_id count
## 1 1 917230
## 2 2 1791695
## 3 3 993116
## 4 77 1123643
```

2.3 Extracting the data from the server

Now we can tell CohortMethod to extract the cohorts, construct covariates, and extract all necessary data for our analysis.

Important: The target and comparator drug must not be included in the covariates, including any descendant concepts. You will need to manually add the drugs and descendants to the `excludedCovariateConceptIds` of the covariate settings. In this example code we exclude the concepts for celecoxib and diclofenac and specify `addDescendantsToExclude = TRUE`:

```
# Define which types of covariates must be constructed:
covSettings <- createDefaultCovariateSettings(
  excludedCovariateConceptIds = c(diclofenacConceptId, celecoxibConceptId),
  addDescendantsToExclude = TRUE
)

#Load data:
cohortMethodData <- getDbCohortMethodData(
  connectionDetails = connectionDetails,
  cdmDatabaseSchema = cdmDatabaseSchema,
  targetId = 1,
  comparatorId = 2,
  outcomeIds = 77,
  exposureDatabaseSchema = cohortDatabaseSchema,
  exposureTable = cohortTable,
  outcomeDatabaseSchema = cohortDatabaseSchema,
  outcomeTable = cohortTable,
  nestingCohortDatabaseSchema = cohortDatabaseSchema,
  nestingCohortTable = cohortTable,
  getDbCohortMethodDataArgs = createGetDbCohortMethodDataArgs(
    removeDuplicateSubjects = "keep first, truncate to second",
    firstExposureOnly = TRUE,
    washoutPeriod = 365,
    restrictToCommonPeriod = TRUE,
  )
)
```

```

    nestingCohortId = 3,
    covariateSettings = covSettings
  )
)
cohortMethodData

## # CovariateData object
##
## All cohorts
##
## Inherits from Andromeda:
## # Andromeda object
## # Physical location: C:\Users\admin_mschuemi.EU\AppData\Local\Temp\2\RtmpspXStT\file41a83f2a574a.duckdb
##
## Tables:
## $analysisRef (analysisId, analysisName, domainId, startDay, endDay, isBinary, missingMeansZero)
## $cohorts (rowId, personSeqId, personId, treatment, cohortStartDate, daysFromObsStart, daysToCohortEnd, daysToObsEnd)
## $covariateRef (covariateId, covariateName, analysisId, conceptId, valueAsConceptId, collisions)
## $covariates (rowId, covariateId, covariateValue)
## $outcomes (rowId, outcomeId, daysToEvent)

```

There are many parameters, but they are all documented in the `CohortMethod` manual. The `createDefaultCovariateSettings` function is described in the `FeatureExtraction` package. In short, we are pointing the function to the table created earlier and indicating which concept IDs in that table identify the target, comparator, nesting cohort and outcome. We instruct that the default set of covariates should be constructed, including covariates for all conditions, drug exposures, and procedures that were found on or before the index date. To customize the set of covariates, please refer to the `FeatureExtraction` package vignette by typing `vignette("UsingFeatureExtraction", package="FeatureExtraction")`.

We let the `CohortMethod` package construct our sets of new users: - For those patients who have exposure to both the target and the comparator, we keep whichever is their first, and truncate their exposure at the start of the second (if the first occurrences of the target and comparator start simultaneously the patient is removed). - We restrict to the first exposure overall (in this case redundant with the previous step). - We require a washout period of 365 days, meaning any patients with less than 365 days of prior observation are removed. - We restrict to the period in time when both drugs were observed in the database. This can be especially helpful when one of the exposures is new to the market. - We restrict to the nesting cohort, in this case our indication. Only exposures within the nesting cohort are kept.

We can see how many persons and exposures were left after each of these steps:

```

getAttritionTable(cohortMethodData)

```

	description	targetPersons	comparatorPersons	targetExposures	comparatorExposures
## 1	Original cohorts	917230	993116	917230	993116
## 2	Keep first, truncate when e ...	845385	873667	845385	873667
## 3	First exposure only	845385	873667	845385	873667
## 4	365 days of prior observati ...	358646	515193	358646	515193
## 5	Restrict to common period	358646	515193	358646	515193
## 6	Restrict to nesting cohort	89318	128301	89318	128301

The `cohortMethodData()` function extracts all data about the exposures, outcomes, and covariates from the server and stores them in the `cohortMethodData` object. This object uses the `Andromeda` package to store information in a way that ensures R does not run out of memory, even when the data are large. We can use the `genericSummary()` function to view some more information of the data we extracted:

```

summary(cohortMethodData)

```

##	Event count	Person count
## 77	37448	20123

2.3.1 Saving the data to file

Creating the `cohortMethodData` file can take considerable computing time, and it is probably a good idea to save it for future sessions. Because `cohortMethodData` uses `Andromeda`, we cannot use R's regular save function. Instead, we'll have to use the `saveCohortMethodData()` function:

```
saveCohortMethodData(cohortMethodData, "coxibVsNonseIvsGiBleed.zip")
```

We can use the `loadCohortMethodData()` function to load the data in a future session.

3 Defining the study population

Typically, the exposure cohorts and outcome cohorts will be defined independently of each other. When we want to produce an effect size estimate, we need to further restrict these cohorts and put them together, for example by removing exposed subjects that had the outcome prior to exposure, and only keeping outcomes that fall within a defined risk window. For this we can use the `createStudyPopulation` function:

```
studyPop <- createStudyPopulation(  
  cohortMethodData = cohortMethodData,  
  outcomeId = 77,  
  createStudyPopulationArgs = createCreateStudyPopulationArgs(  
    removeSubjectsWithPriorOutcome = TRUE,  
    priorOutcomeLookback = 365,  
    minDaysAtRisk = 1,  
    riskWindowStart = 0,  
    startAnchor = "cohort start",  
    riskWindowEnd = 30,  
    endAnchor = "cohort end"  
  )  
)
```

We specify the outcome ID we will use, and that people who had the outcomes in the 365 days before the risk window start date will be removed. The risk window is defined as starting at the cohort start date (the index date, `riskWindowStart = 0` and `startAnchor = "cohort start"`), and the risk windows ends 30 days after the cohort ends (`riskWindowEnd = 30` and `endAnchor = "cohort end"`). Note that the risk windows are truncated at the end of observation or the study end date. We also remove subjects who have no time at risk. To see how many people are left in the study population we can always use the `getAttritionTable` function:

```
getAttritionTable(studyPop)
```

##	description	targetPersons	comparatorPersons	targetExposures	comparatorExposures
## 1	Original cohorts	917230	993116	917230	993116
## 2	Keep first, truncate when e ...	845385	873667	845385	873667
## 3	First exposure only	845385	873667	845385	873667
## 4	365 days of prior observati ...	358646	515193	358646	515193
## 5	Restrict to common period	358646	515193	358646	515193
## 6	Restrict to nesting cohort	89318	128301	89318	128301
## 7	No prior outcome	88134	126316	88134	126316
## 8	Have at least 1 days at ris ...	88134	126316	88134	126316

4 Propensity scores

The `CohortMethod` can use propensity scores to adjust for potential confounders. Instead of the traditional approach of using a handful of predefined covariates, `CohortMethod` typically uses tens of thousands of covariates that are automatically constructed based on conditions, procedures and drugs in the records of the subjects.

4.1 Fitting a propensity model

We can fit a propensity model using the covariates constructed by the `getDbcohortMethodData()` function:

```
ps <- createPs(cohortMethodData = cohortMethodData, population = studyPop)
```

The `createPs()` function uses the `Cyclops` package to fit a large-scale regularized logistic regression.

To fit the propensity model, `Cyclops` needs to know the hyperparameter value which specifies the variance of the prior. By default `Cyclops` will use cross-validation to estimate the optimal hyperparameter. However, be aware that this can take a really long time. You can use the `prior` and `control` parameters of the `createPs()` to specify `Cyclops` behavior, including using multiple CPUs to speed-up the cross-validation.

4.2 Propensity score diagnostics

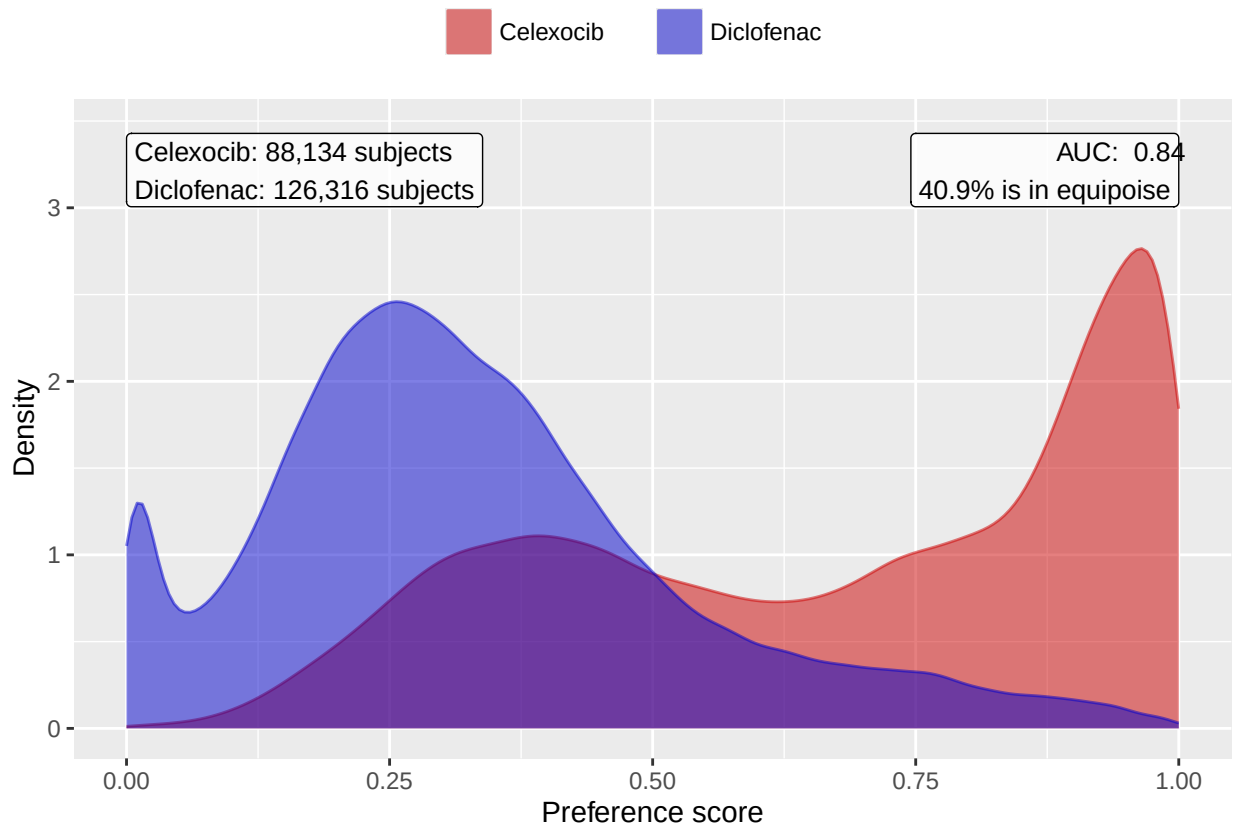
We can compute the area under the receiver-operator curve (AUC) for the propensity score model:

```
computePsAuc(ps)
```

```
## [1] 0.8411435
```

We can also plot the propensity score distribution. By default, the `plotPS()` function shows the preference score, a transformation of the propensity score that adjusts for differences in sizes between the target and comparator:

```
plotPS(ps,  
  targetLabel = "Celexocib",  
  comparatorLabel = "Diclofenac",  
  showCountsLabel = TRUE,  
  showAucLabel = TRUE,  
  showEquipoiseLabel = TRUE)
```



It is also possible to inspect the propensity model itself by showing the covariates that have non-zero coefficients:

```
getPsModel(ps, cohortMethodData)
```

```
## # A tibble: 6 x 3
##   coefficient covariateId covariateName
##   <dbl>      <dbl> <chr>
## 1    -4.11  1150871413 ...gh 0 days relative to index: misoprostol
## 2     3.26   2001006 index year: 2001
## 3     3.14   2002006 index year: 2002
## 4     2.67   2003006 index year: 2003
## 5     2.38   2004006 index year: 2004
## 6     1.67   2007006 index year: 2007
```

One advantage of using the regularization when fitting the propensity model is that most coefficients will shrink to zero and fall out of the model. It is a good idea to inspect the remaining variables for anything that should not be there, for example variations of the drugs of interest that we forgot to exclude.

Finally, we can inspect the percent of the population in equipoise, meaning they have a preference score between 0.3 and 0.7:

```
CohortMethod::computeEquipoise(ps)
```

```
## [1] 0.4090277
```

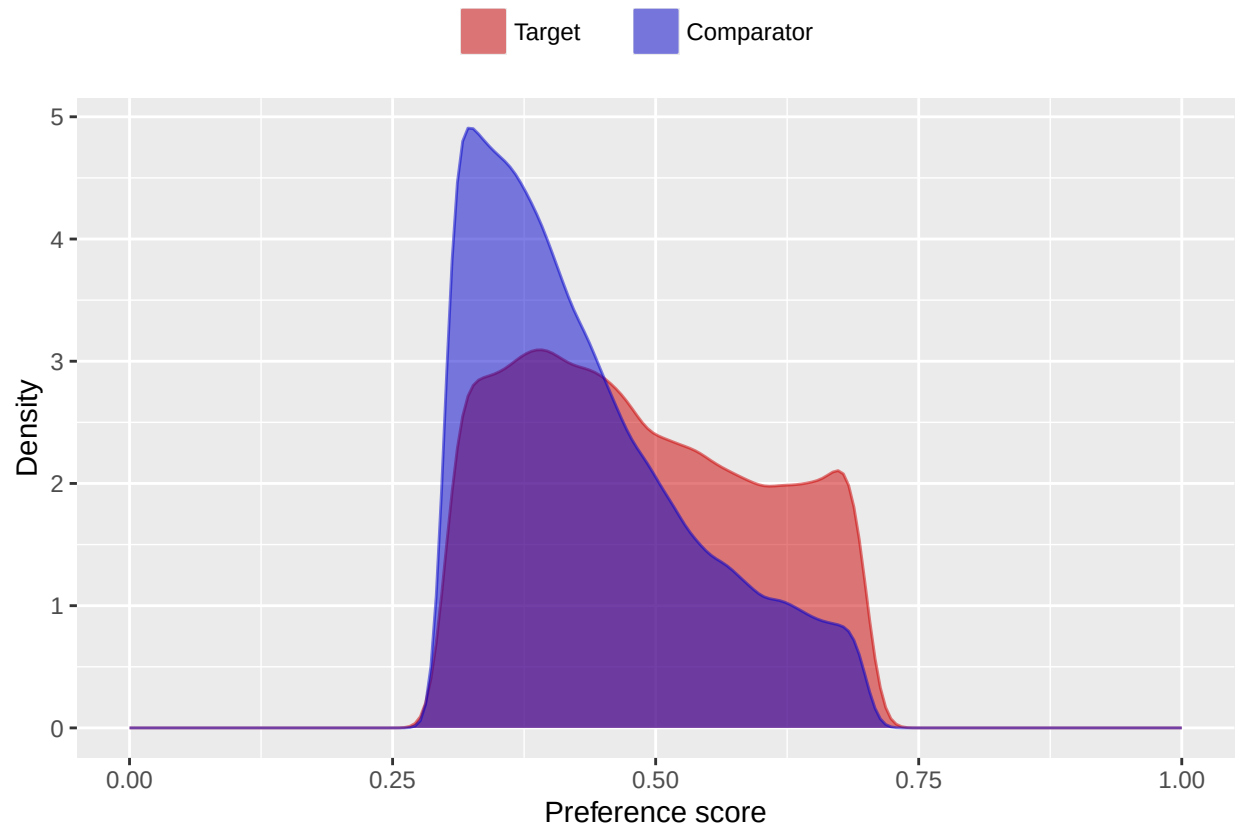
A low equipoise indicates there is little overlap between the target and comparator populations.

4.3 Using the propensity score

We can use the propensity scores to trim, stratify, match, or weigh our population. For example, one could trim to equipoise, meaning only subjects with a preference score between 0.3 and 0.7 are kept:

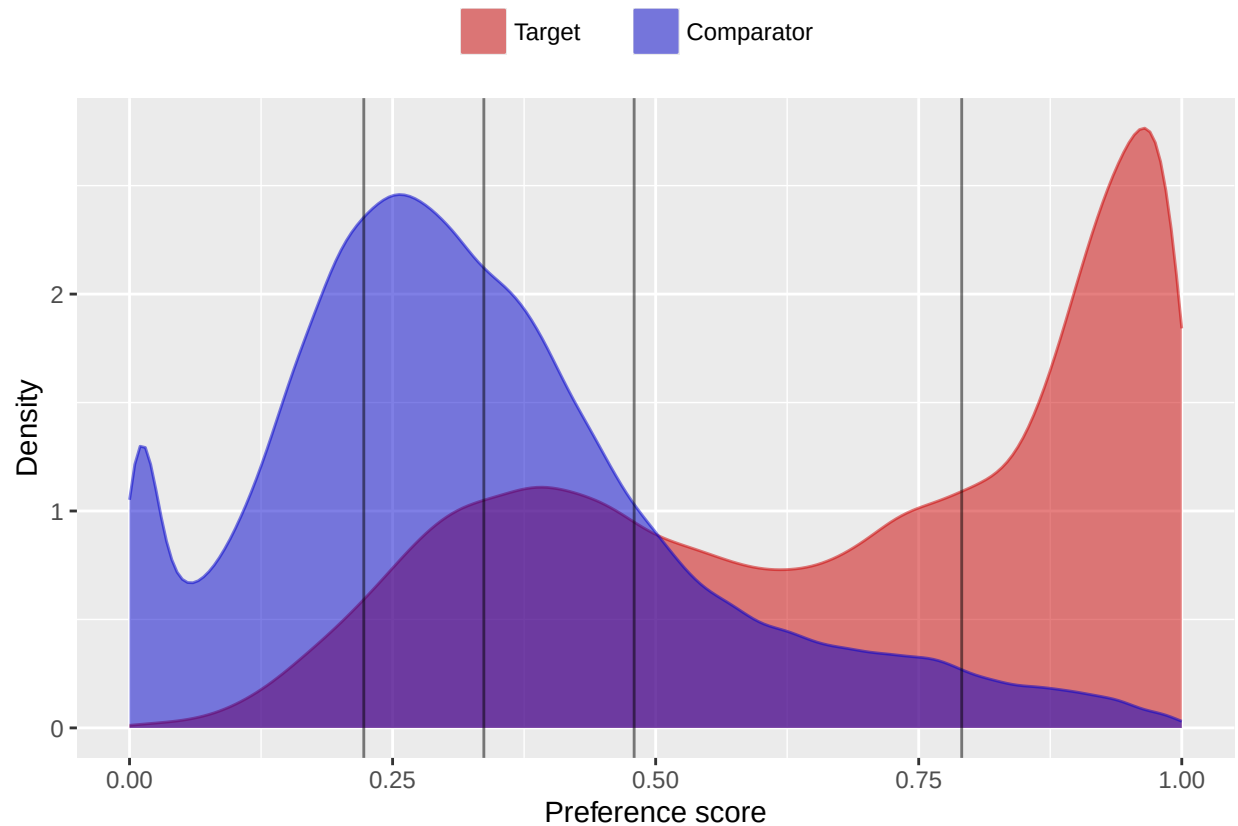
```
trimmedPop <- trimByPs(ps,
  trimByPsArgs = createTrimByPsArgs(
    equipoiseBounds = c(0.3, 0.7)
  ))
# Note: we need to also provide the original PS object so the preference score
# is computed using the original relative sizes of the cohorts:
plotPs(trimmedPop, ps)
```

```
## Trimming removed 55955 (63.5%) rows from the target, 70779 (56.0%) rows from the comparator in total
```

Instead (or additionally), we could stratify the population based on the propensity score:

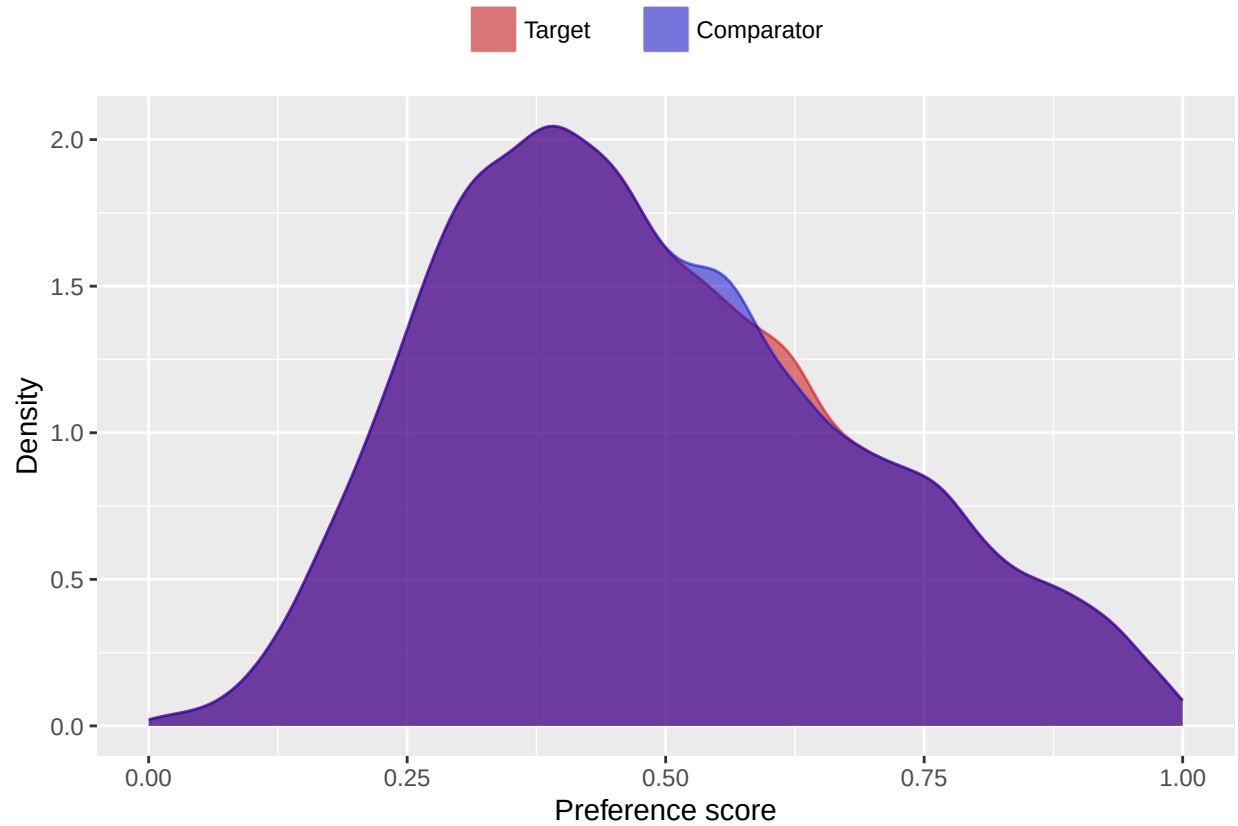
```
stratifiedPop <- stratifyByPs(ps,  
  stratifyByPsArgs = createStratifyByPsArgs(  
    numberOfStrata = 5  
  ))  
plotPs(stratifiedPop)
```



We can also match subjects based on propensity scores. In this example, we're using one-to-one matching. By default, `createPs()` will use a caliper of 0.2 on the standardized logit scale:

```
matchedPop <- matchOnPs(ps,
  matchOnPsArgs = createMatchOnPsArgs(
    maxRatio = 1
  ))
plotPs(matchedPop, ps)
```

Population size after matching is 96046



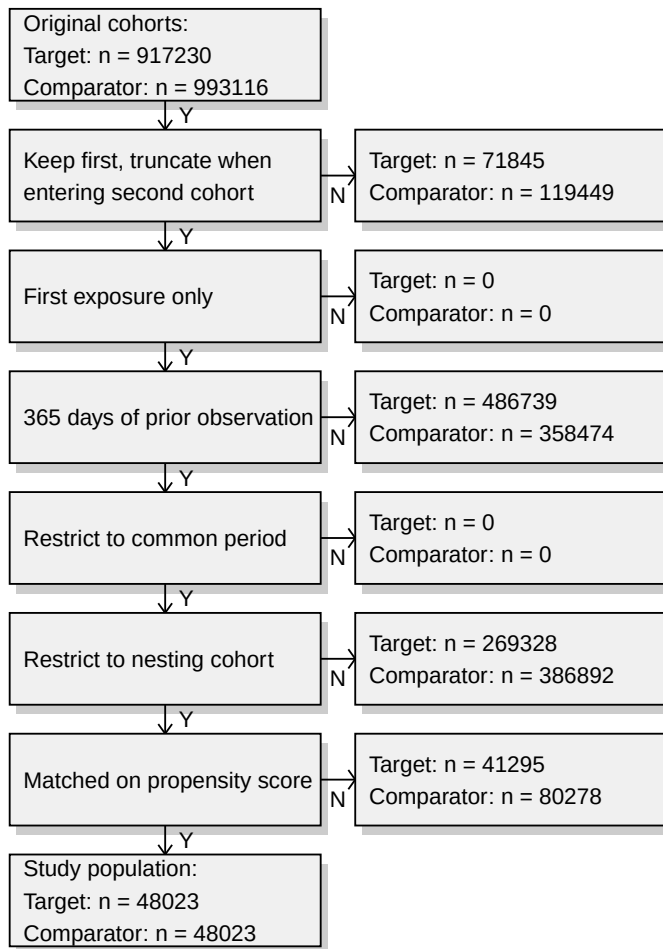
We can see the effect of trimming and/or matching on the population using the `getAttritionTable` function:

```
getAttritionTable(matchedPop)
```

##	description	targetPersons	comparatorPersons	targetExposures	comparatorExposures
## 1	Original cohorts	917230	993116	917230	993116
## 2	Keep first, truncate when e ...	845385	873667	845385	873667
## 3	First exposure only	845385	873667	845385	873667
## 4	365 days of prior observati ...	358646	515193	358646	515193
## 5	Restrict to common period	358646	515193	358646	515193
## 6	Restrict to nesting cohort	89318	128301	89318	128301
## 7	Matched on propensity score	48023	48023	48023	48023

Or, if we like, we can plot an attrition diagram:

```
drawAttritionDiagram(matchedPop)
```

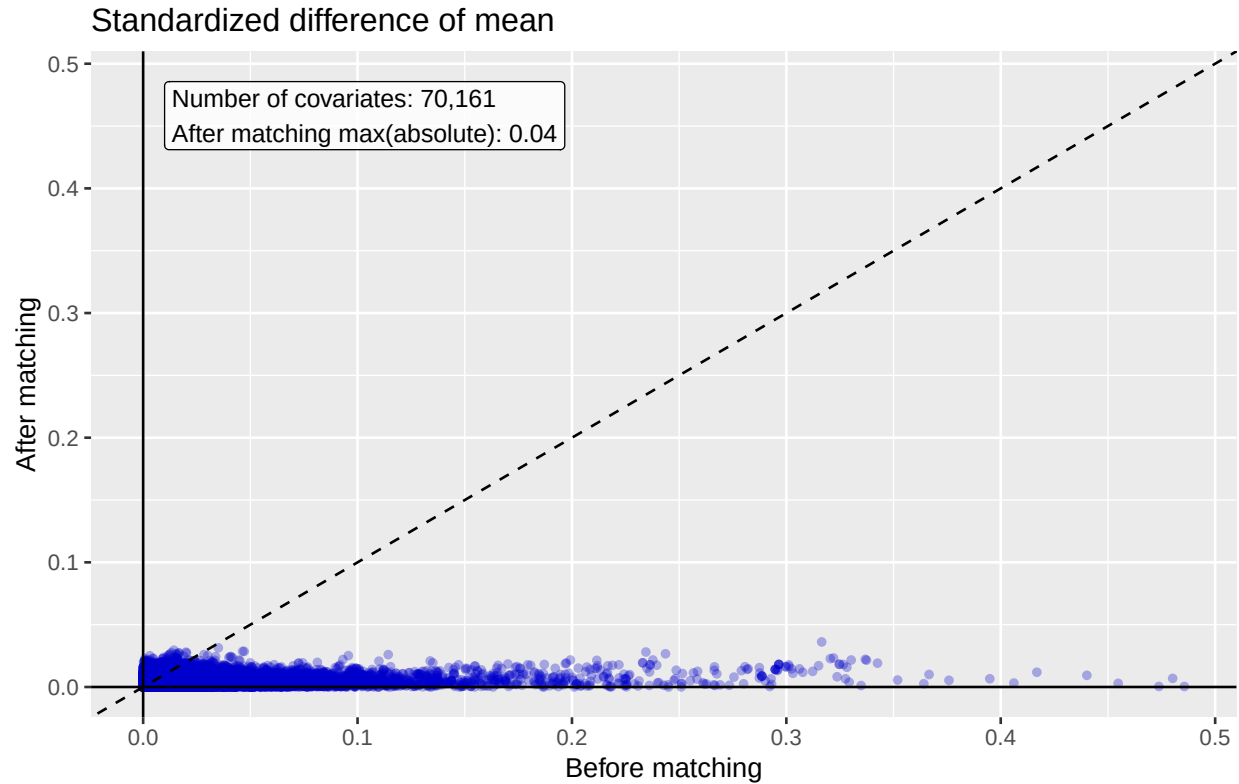


4.4 Evaluating covariate balance

To evaluate whether our use of the propensity score is indeed making the two cohorts more comparable, we can compute the covariate balance before and after trimming, matching, and/or stratifying:

```
balance <- computeCovariateBalance(matchedPop, cohortMethodData)
```

```
plotCovariateBalanceScatterPlot(balance,
                                showCovariateCountLabel = TRUE,
                                showMaxLabel = TRUE)
```



```
plotCovariateBalanceOfTopVariables(balance)
```



The 'before matching' population is the population as extracted by the `getDbCohortMethodData` function, so

before any further filtering steps. We typically consider the populations to be balanced if, after PS adjustment, all covariates have a standardized difference of means smaller than 0.1.

4.5 Inspecting select population characteristics

It is customary to include a table in your paper that lists some select population characteristics before and after matching/stratification/trimming. This is usually the first table, and so will be referred to as ‘table 1’. To generate this table, you can use the `createCmTable1` function:

```
createCmTable1(balance)
```

Characteristic	Before matching			After matching		
	Target %	Comparator %	Std. diff	Target %	Comparator %	Std. diff
Age group						
40 - 44	0.0	0.0	0.00	0.0	0.0	0.01
45 - 49	0.0	0.0	0.00	0.0	0.0	0.00
50 - 54	0.1	0.1	0.00	0.1	0.2	-0.01
55 - 59	0.4	0.5	-0.02	0.4	0.5	-0.01
60 - 64	0.9	1.3	-0.04	1.1	1.1	-0.01
65 - 69	22.9	21.2	0.04	22.5	22.4	0.00
70 - 74	28.3	26.6	0.04	27.3	27.2	0.00
75 - 79	22.0	20.9	0.03	21.3	21.5	-0.01
80 - 84	14.9	15.3	-0.01	15.2	15.1	0.00
85 - 89	7.4	9.3	-0.07	8.4	8.2	0.01
90 - 94	2.5	3.8	-0.07	3.0	3.0	0.00
95 - 99	0.5	0.9	-0.06	0.6	0.6	0.00
100 - 104	0.0	0.1	-0.02	0.1	0.1	0.00
105 - 109	0.0	0.0	-0.01	0.0	0.0	-0.01
Gender: female	62.9	67.2	-0.09	65.2	65.0	0.01
Medical history: General						
Acute respiratory disease	21.7	25.4	-0.09	23.6	23.5	0.00
Attention deficit hyperactivity disorder	0.2	0.2	0.00	0.2	0.2	0.00
Chronic liver disease	1.0	1.5	-0.05	1.1	1.1	-0.01
Chronic obstructive pulmonary disease	9.6	11.3	-0.05	10.0	9.9	0.00
Crohn's disease	0.3	0.4	-0.02	0.4	0.4	0.00
Dementia	2.5	4.0	-0.08	3.0	2.9	0.00
Depressive disorder	9.1	11.9	-0.09	10.2	10.1	0.00
Diabetes mellitus	22.7	28.4	-0.13	24.7	24.6	0.00
Gastroesophageal reflux disease	16.1	19.4	-0.09	17.4	17.0	0.01
Gastrointestinal hemorrhage	3.5	3.8	-0.02	2.2	2.4	-0.01
Human immunodeficiency virus infection	0.0	0.1	-0.02	0.1	0.0	0.00
Hyperlipidemia	45.5	55.3	-0.20	49.6	48.6	0.02
Hypertensive disorder	63.2	69.7	-0.14	65.3	65.0	0.01
Lesion of liver	0.5	0.8	-0.04	0.5	0.5	0.00
Obesity	10.2	11.7	-0.05	10.0	9.7	0.01
Osteoarthritis	82.3	76.0	0.15	77.9	78.1	-0.01
Pneumonia	4.3	5.2	-0.05	4.3	4.3	0.00
Psoriasis	1.3	1.5	-0.02	1.4	1.3	0.01
Renal impairment	6.9	13.4	-0.22	8.3	8.1	0.01
Rheumatoid arthritis	3.1	3.9	-0.05	3.4	3.4	0.00
Schizophrenia	0.1	0.1	-0.01	0.1	0.1	0.00
Ulcerative colitis	0.5	0.6	-0.01	0.5	0.5	0.00
Urinary tract infectious disease	11.0	13.4	-0.08	11.8	11.8	0.00
Viral hepatitis C	0.2	0.3	-0.02	0.2	0.2	0.00
Medical history: Cardiovascular disease						
Atrial fibrillation	9.8	11.8	-0.07	10.0	10.0	0.00
Cerebrovascular disease	11.2	12.8	-0.05	11.5	11.5	0.00
Coronary arteriosclerosis	19.2	20.9	-0.04	18.9	19.1	-0.01
Heart disease	43.4	44.8	-0.03	41.9	41.9	0.00
Heart failure	7.9	10.4	-0.08	7.8	7.9	0.00
Ischemic heart disease	9.1	9.6	-0.02	8.4	8.5	0.00
Peripheral vascular disease	8.8	12.5	-0.12	9.8	9.8	0.00
Pulmonary embolism	1.1	1.2	-0.02	1.1	1.1	0.00
Venous thrombosis	3.3	3.9	-0.03	3.4	3.6	-0.01
Medical history: Neoplasms						
Malignant lymphoma	0.7	0.8	-0.01	0.8	0.8	0.00
Malignant neoplasm of anorectum	0.3	0.3	0.01	0.3	0.3	0.00
Malignant neoplastic disease	18.6	19.3	-0.02	19.0	19.1	0.00
Malignant tumor of breast	3.7	4.1	-0.02	3.9	3.9	0.00
Malignant tumor of colon	0.7	0.7	0.00	0.7	0.7	0.00
Malignant neoplasm of lung	0.3	0.4	-0.02	0.4	0.4	0.00
Malignant neoplasm of urinary bladder	0.9	0.8	0.00	0.9	0.8	0.01
Primary malignant neoplasm of prostate	3.8	3.3	0.02	3.6	3.6	0.00
Medication use						
Agents acting on the renin-angiotensin system	50.1	53.4	-0.07	52.2	52.0	0.00
Antibacterials for systemic use	68.8	70.5	-0.04	68.6	68.6	0.00
Antidepressants	28.2	30.3	-0.05	30.0	30.1	0.00
Antiepileptics	20.0	21.6	-0.04	20.6	20.9	-0.01
Antiinflammatory and antirheumatic products	36.4	36.1	0.01	37.3	37.3	0.00
Antineoplastic agents	4.9	5.8	-0.04	5.2	5.1	0.00
Antipsoriatics	0.9	1.3	-0.05	0.9	1.0	-0.01
Antithrombotic agents	29.6	24.4	0.12	24.4	24.6	-0.01
Beta blocking agents	36.9	41.2	-0.09	38.3	38.0	0.01
Calcium channel blockers	28.1	31.4	-0.07	29.3	29.1	0.00
Diuretics	45.0	46.6	-0.03	45.2	45.4	0.00
Drugs for acid related disorders	41.1	43.1	-0.04	39.8	40.1	-0.01
Drugs for obstructive airway diseases	51.1	54.6	-0.07	53.6	53.0	0.01
Drugs used in diabetes	18.4	22.1	-0.09	19.6	19.5	0.00
Immunosuppressants	4.1	5.8	-0.08	4.8	4.6	0.01
Lipid modifying agents	55.2	59.2	-0.08	57.9	57.5	0.01
Opioids	64.7	55.1	0.20	58.5	59.0	-0.01

Psycholeptics	34.1	32.8	0.03	33.8	34.2	-0.01
Psychostimulants, agents used for adhd and nootropics	1.7	1.9	-0.02	2.0	2.1	-0.01

4.6 Generalizability

The goal of any propensity score adjustments is typically to make the target and comparator cohorts comparably, to allow proper causal inference. However, in doing so, we often need to modify our population, for example dropping subjects that have no counterpart in the other exposure cohort. The population we end up estimating an effect for may end up being very different from the population we started with. An important question is: how different? And in what ways? If the populations before and after adjustment are very different, our estimated effect may not generalize to the original population (if effect modification is present). The `getGeneralizabilityTable()` function informs on these differences:

```
getGeneralizabilityTable(balance)

# A tibble: 85,838 x 5
  covariateId covariateName      beforeMatchingMean afterMatchingMean stdDiff
  <dbl> <chr>                <dbl>                <dbl> <dbl>
1  4160439504 ...: Administration of anesthesia      0.157      0.0299  0.447
2  2105103504 ...cing (total knee arthroplasty)      0.116      0.0102  0.446
3  38003162804 ...s and Devices - Other Implants      0.124      0.0185  0.420
4  38003208804 ...vices - General Classification      0.139      0.0296  0.401
5  38003390804 ... Room - General Classification      0.132      0.0279  0.391
6  38003213804 ...hesia - General Classification      0.120      0.0247  0.374
7   764608504 ...dex: Preprocedural examination      0.0978      0.0156  0.361
8  38003245804 ... - Evaluation Or Re-Evaluation      0.118      0.0302  0.339
9  38003138804 ...rmacy - General Classification      0.175      0.0692  0.326
10 4002014212 ...ndex: Pain following procedure      0.0753      0.0110  0.321
# i 85,828 more rows
```

In this case, because we used PS matching, we are likely aiming to estimate the average treatment effect in the treated (ATT). For this reason, the `getGeneralizabilityTable()` function automatically selected the target cohort as the basis for evaluating generalizability: it shows, for each covariate, the mean value before and PS adjustment in the target cohort. Also shown is the standardized difference of mean, and the table is reverse sorted by the absolute standard difference of mean (ASDM).

5 Follow-up and power

Before we start fitting an outcome model, we might be interested to know whether we have sufficient power to detect a particular effect size. It makes sense to perform these power calculations once the study population has been fully defined, so taking into account loss to the various inclusion and exclusion criteria (such as no prior outcomes), and loss due to matching and/or trimming. Since the sample size is fixed in retrospective studies (the data has already been collected), and the true effect size is unknown, the CohortMethod package provides a function to compute the minimum detectable relative risk (MDRR) instead:

```
computeMdr(
  population = studyPop,
  modelType = "cox",
  alpha = 0.05,
  power = 0.8,
  twoSided = TRUE
)

## targetPersons comparatorPersons targetExposures comparatorExposures targetDays comparatorDays totalOutcomes mdr se
## 1 88134 126316 88134 126316 12087401 9619724 1252 1.174598 0.05744113
```

In this example we used the `studyPop` object, so the population before any matching or trimming. If we want to know the MDRR after matching, we use the `matchedPop` object we created earlier instead:

```
computeMdr(
  population = matchedPop,
  modelType = "cox",
  alpha = 0.05,
  power = 0.8,
  twoSided = TRUE
)
```

```
##   targetPersons comparatorPersons targetExposures comparatorExposures targetDays comparatorDays totalOutcomes   mdr   se
## 1         48023          48023         48023          48023         6752916        3896834          597 1.257748 0.08185455
```

Even though the MDRR in the matched population is higher, meaning we have less power, we should of course not be fooled: matching most likely eliminates confounding, and is therefore preferred to not matching.

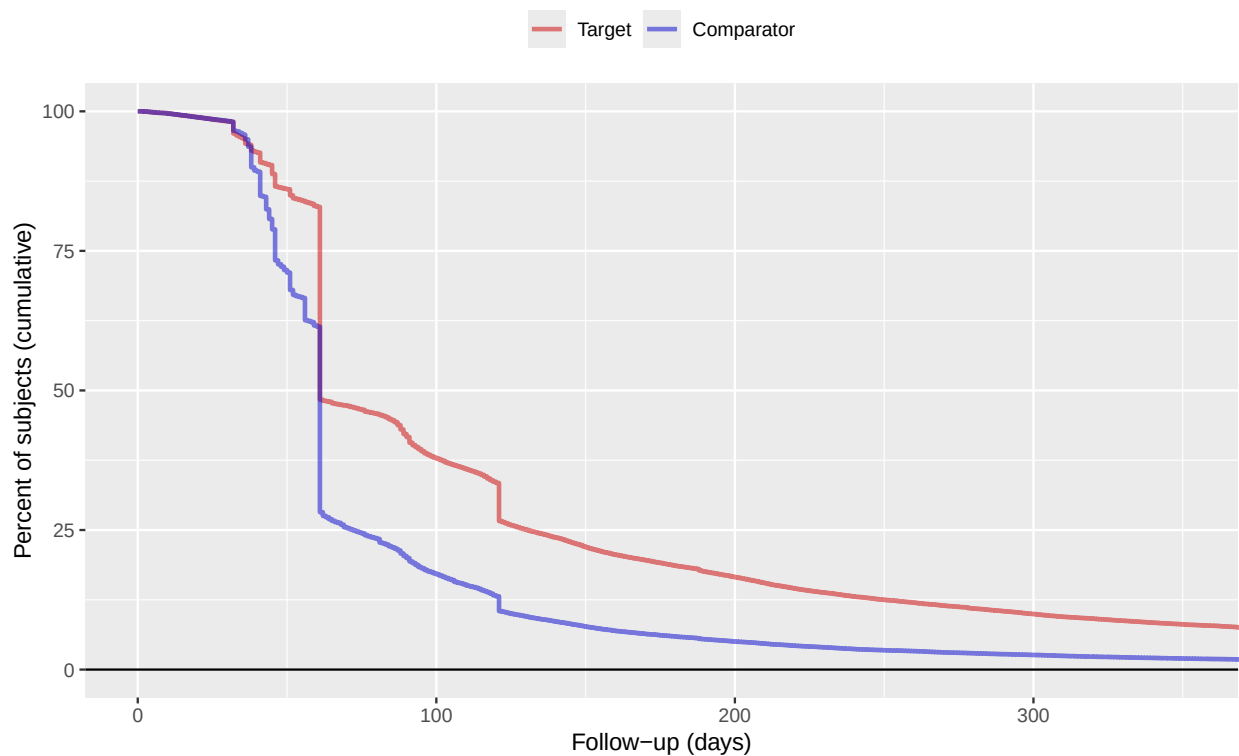
To gain a better understanding of the amount of follow-up available we can also inspect the distribution of follow-up time. We defined follow-up time as time at risk, so not censored by the occurrence of the outcome. The `getFollowUpDistribution` can provide a simple overview:

```
getFollowUpDistribution(population = matchedPop)
```

```
##   100% 75% 50% 25% 0% Treatment
## 1     1  60  60 130 3486         1
## 2     1  45  60  71 3833         0
```

The output is telling us number of days of follow-up each quantile of the study population has. We can also plot the distribution:

```
plotFollowUpDistribution(population = matchedPop)
```



6 Outcome models

The outcome model is a model describing which variables are associated with the outcome.

6.1 Fitting a simple outcome model

In theory we could fit an outcome model without using the propensity scores. In this example we are fitting an outcome model using a Cox regression:

```
outcomeModel <- fitOutcomeModel(  
  population = studyPop,  
  fitOutcomeModelArgs = createFitOutcomeModelArgs(  
    modelType = "cox"  
  )  
)  
outcomeModel  
  
##           Estimate lower .95 upper .95      logRr seLogRr  
## treatment  0.961778  0.856018  1.080701 -0.038971  0.0595
```

But of course we want to make use of the matching done on the propensity score:

```
outcomeModel <- fitOutcomeModel(  
  population = matchedPop,  
  fitOutcomeModelArgs = createFitOutcomeModelArgs(  
    modelType = "cox"  
  )  
)  
outcomeModel  
  
##           Estimate lower .95 upper .95      logRr seLogRr  
## treatment  0.84469   0.71412  1.00030 -0.16878   0.086
```

Note that we define the sub-population to be only those in the `matchedPop` object, which we created earlier by matching on the propensity score.

Instead of matching or stratifying we can also perform Inverse Probability of Treatment Weighting (IPTW):

```
outcomeModel <- fitOutcomeModel(  
  population = ps,  
  fitOutcomeModelArgs = createFitOutcomeModelArgs(  
    modelType = "cox",  
    inversePtWeighting = TRUE,  
    bootstrapCi = TRUE  
  )  
)  
outcomeModel  
  
##           Estimate lower .95 upper .95      logRr seLogRr  
## treatment  0.66899   0.47331  0.97783 -0.40199   0.178
```

Note that in this case we may want to use a bootstrap to compute the confidence interval.

6.2 Adding interaction terms

We may be interested whether the effect is different across different groups in the population. To explore this, we may include interaction terms in the model. In this example we include three interaction terms:

```
interactionCovariateIds <- c(8532001, 201826210, 21600960413)  
# 8532001 = Female  
# 201826210 = Type 2 Diabetes  
# 21600960413 = Concurrent use of antithrombotic agents  
outcomeModel <- fitOutcomeModel(  
  population = matchedPop,  
  fitOutcomeModelArgs = createFitOutcomeModelArgs(  
    modelType = "cox",  
    interactionCovariateIds = interactionCovariateIds,  
    bootstrapCi = TRUE  
  )  
)  
outcomeModel
```

```

cohortMethodData = cohortMethodData,
fitOutcomeModelArgs = createFitOutcomeModelArgs(
  modelType = "cox",
  interactionCovariateIds = interactionCovariateIds
)
)
outcomeModel

```

```

##
## treatment
## treatment * condition_era group (ConditionGroupEraLongTerm) during day -365 through 0 days relative to index: Type 2 diabetes mellitus
## treatment * drug_era group (DrugGroupEraOverlapping) during day 0 through 0 days relative to index: ANTITHROMBOTIC AGENTS
## treatment * gender = FEMALE

```

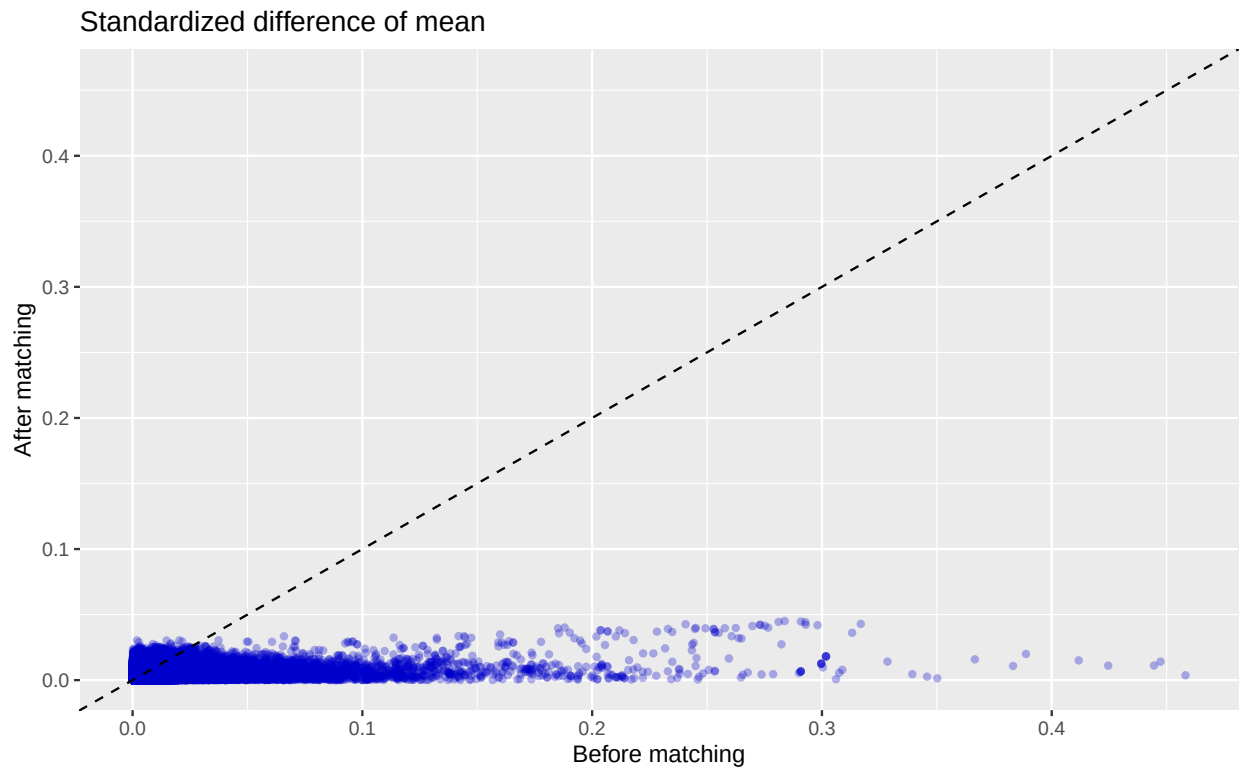
	Estimate	lower .95	upper .95	logRr	seLogRr
treatment	0.809658	0.590267	1.113710	-0.211144	0.1620
treatment * condition_era group (ConditionGroupEraLongTerm) during day -365 through 0 days relative to index: Type 2 diabetes mellitus	0.860438	0.601425	1.233185	-0.150314	0.1832
treatment * drug_era group (DrugGroupEraOverlapping) during day 0 through 0 days relative to index: ANTITHROMBOTIC AGENTS	1.011228	0.701494	1.462634	0.011166	0.1875
treatment * gender = FEMALE	1.111761	0.791320	1.560599	0.105946	0.1732

It is prudent to verify that covariate balance has also been achieved in the subgroups of interest. For example, we can check the covariate balance in the subpopulation of females:

```

balanceFemale <- computeCovariateBalance(
  population,
  cohortMethodData,
  computeCovariateBalanceArgs = createComputeCovariateBalanceArgs(
    subgroupCovariateId = 8532001
  )
)
plotCovariateBalanceScatterPlot(balanceFemale)

```



6.3 Adding covariates to the outcome model

One final refinement would be to use the same covariates we used to fit the propensity model to also fit the outcome model. This way we are more robust against misspecification of the model, and more likely to remove bias. For this we use the regularized Cox regression in the `Cyclops` package. (Note that the treatment variable is automatically excluded from regularization.)

```

outcomeModel <- fitOutcomeModel(
  population = matchedPop,
  cohortMethodData = cohortMethodData,
  fitOutcomeModelArgs = createFitOutcomeModelArgs(
    modelType = "cox",
    useCovariates = TRUE,
  )
)
outcomeModel

##           Estimate lower .95 upper .95      logRr seLogRr
## treatment  0.925449  0.717305  1.194886 -0.077476  0.1302

```

6.4 Inspecting the outcome model

We can inspect more details of the outcome model:

```

exp(coef(outcomeModel))

## 4393168283078717
##           0.9254494

exp(confint(outcomeModel))

## [1] 0.7173049 1.1948858

```

We can also see the covariates that ended up in the outcome model:

```

getOutcomeModel(outcomeModel, cohortMethodData)

##   coefficient      id      name
## 1  -0.0774758 4.393168e+15 Treatment

```

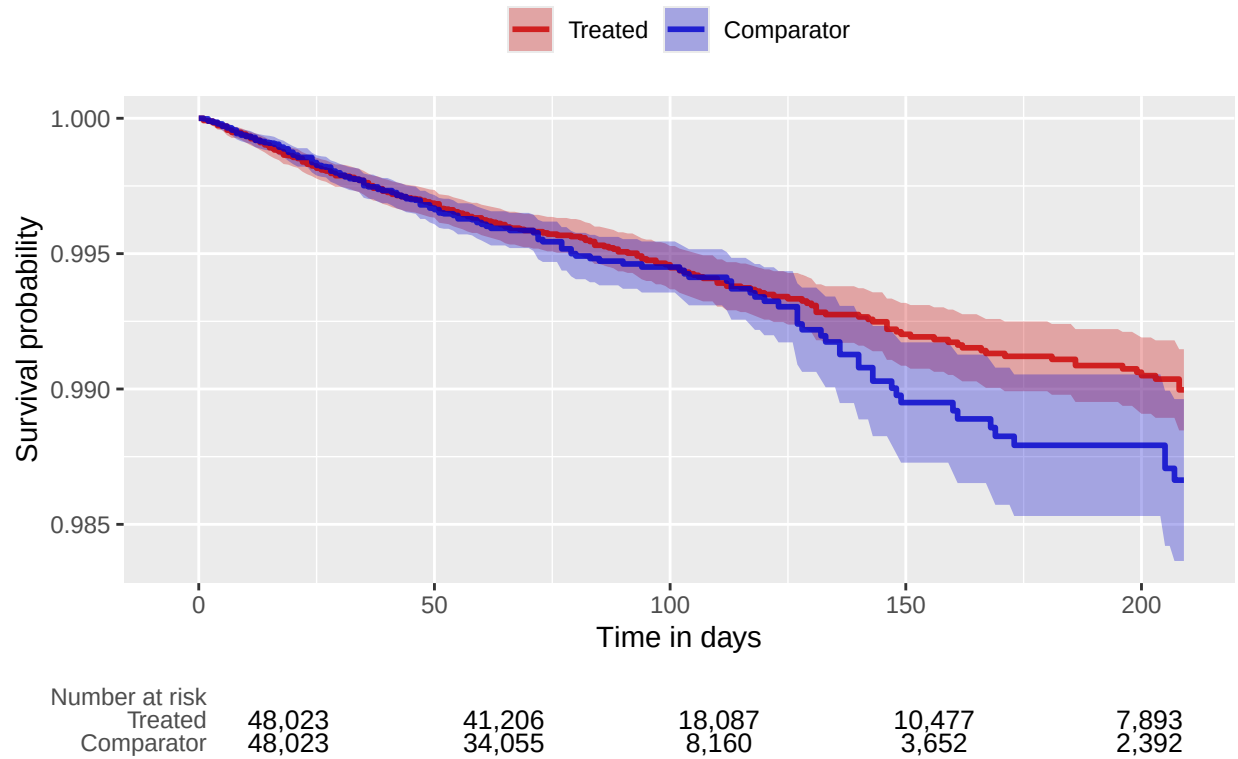
6.5 Kaplan-Meier plot

We can create the Kaplan-Meier plot:

```

plotKaplanMeier(matchedPop, includeZero = FALSE)

```

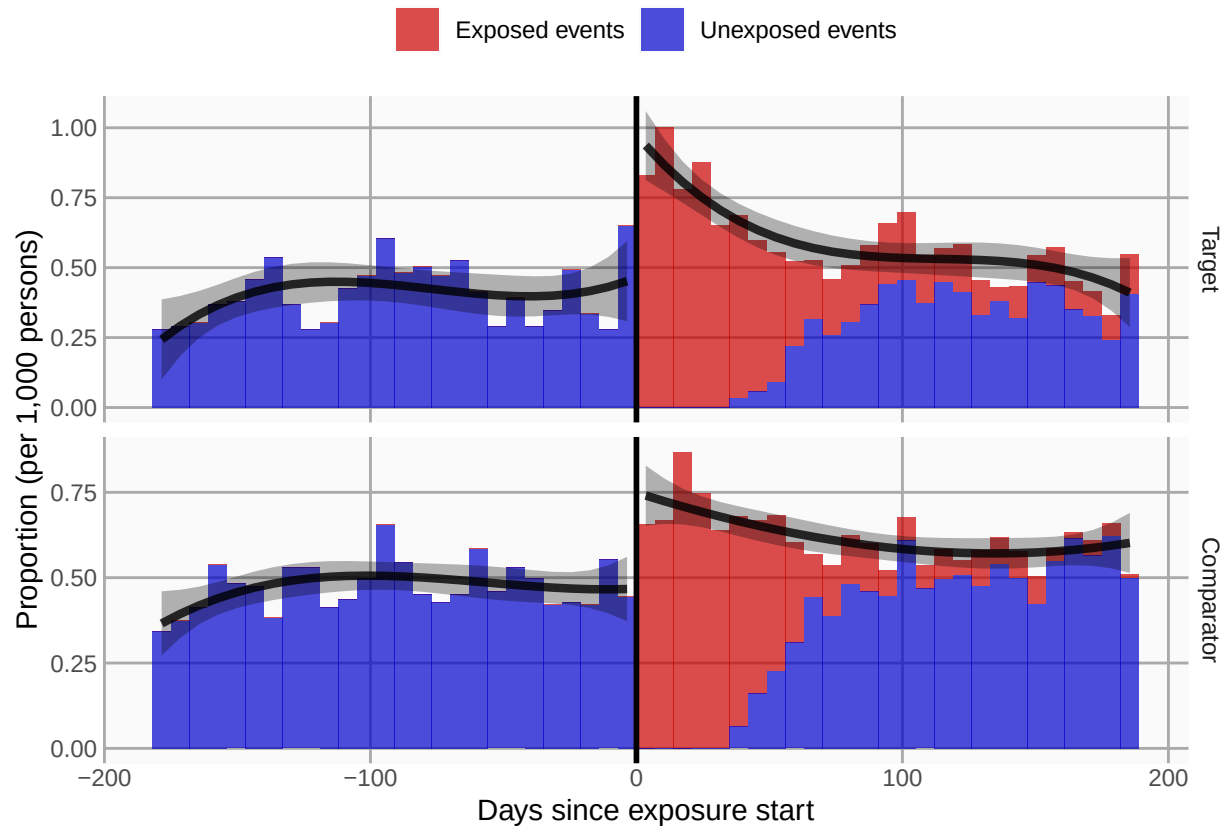


Note that the Kaplan-Meier plot will automatically adjust for any stratification, matching, or trimming that may have been applied.

6.6 Time-to-event plot

We can also plot time-to-event, showing both events before and after the index date, and events during and outside the defined time-at-risk window. This plot can provide insight into the temporal pattern of the outcome relative to the exposures:

```
plotTimeToEvent(
  cohortMethodData = cohortMethodData,
  outcomeId = 77,
  minDaysAtRisk = 1,
  riskWindowStart = 0,
  startAnchor = "cohort start",
  riskWindowEnd = 30,
  endAnchor = "cohort end"
)
```



Note that this plot does not show any adjustment for the propensity score.

7 Acknowledgments

Considerable work has been dedicated to provide the CohortMethod package.

```
citation("CohortMethod")
```

```
## To cite CohortMethod in publications use:
##
##   Schuemie MJ, Reps JM, Black A, DeFalco F, Evans L, Fridgeirsson E, Gilbert JP, Knoll C, Lavallee M
##   MA (2024). "Health-analytics data to evidence suite (HADES): open-source software for observational
##   966-970. doi:10.3233/SHTI231108 <https://doi.org/10.3233/SHTI231108>, <https://doi.org/10.3233/sht
##
## A BibTeX entry for LaTeX users is
##
##   @Article{
##     title = {Health-analytics data to evidence suite (HADES): open-source software for observational
##     author = {M. J. Schuemie and J. M. Reps and A. Black and F. DeFalco and L. Evans and E. Fridgeirsson
##     journal = {Studies in Health Technology and Informatics},
##     year = {2024},
##     volume = {310},
##     pages = {966-970},
##     doi = {10.3233/SHTI231108},
##     url = {https://doi.org/10.3233/shti231108},
##   }
```

Further, CohortMethod makes extensive use of the Cyclops package.

```
citation("Cyclops")
```

```
## To cite Cyclops in publications use:
```

```
##
```

```
## Suchard MA, Simpson SE, Zorych I, Ryan P, Madigan D (2013). "Massive parallelization of serial inference algorithms for complex generalized linear models".
```

```
## Transactions on Modeling and Computer Simulation, 23(1), 10. doi:10.1145/2414416.2414791 <https://doi.org/10.1145/2414416.2414791>
```

```
##
```

```
## A BibTeX entry for LaTeX users is
```

```
##
```

```
## @Article{,
```

```
##   author = {M. A. Suchard and S. E. Simpson and I. Zorych and P. Ryan and D. Madigan},
```

```
##   title = {Massive parallelization of serial inference algorithms for complex generalized linear models},
```

```
##   journal = {ACM Transactions on Modeling and Computer Simulation},
```

```
##   volume = {23},
```

```
##   pages = {10},
```

```
##   year = {2013},
```

```
##   doi = {10.1145/2414416.2414791},
```

```
## }
```

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